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----- Active space -----

LIMBIC SYSTEM AND HYPOTHALAMUS

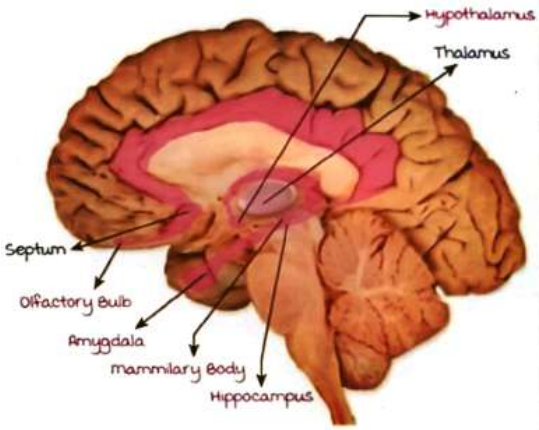
Limbic System

00:01:30

STRUCTURE :

Limbic cortex :

- AKA smell brain/rhinencephalon.
- Rim of cortical tissue.
- Oldest part of human body in terms of evolution.
- Functions : Emotions, smell & memory.



Amygdala :

- Window of limbic system.
- Function : Genesis of emotions/seat of emotions.

Note : Emotions regulated by frontal cortex.

CLINICAL ASPECT :

Lesion at

Bitemporal region, amygdala

Klüver-Bucy syndrome

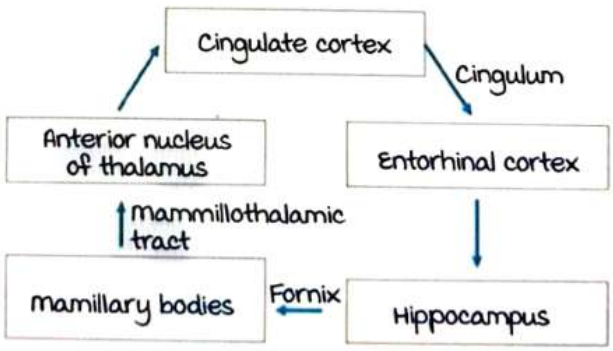
- Visual agnosia (Objects not recognized on sight).
- Hypermetamorphosis (Compulsion to explore environment).
- Placidity (No fears/anger).
- Hyperorality/hyperphagia (Indiscriminate habit of eating).
- Hypersexuality.

Medial temporal lobe

Mesial temporal lobe epilepsy

- Complex partial seizure.
- Uncinate fits (Smell/taste hallucination).
- Deja vu (Unfamiliar environment seems familiar).
- Jamais vu (Familiar environment seems unfamiliar).

PAPEZ CIRCUIT :



Connects major functions (memory & emotion) of limbic system.

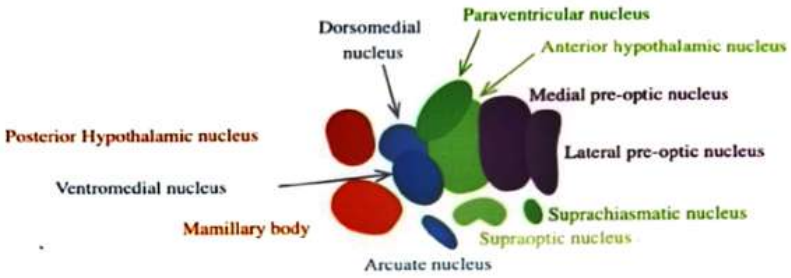
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Hypothalamus

00:16:42 ----- Active space -----

Head ganglion of autonomic nervous System.



FUNCTIONS :

1. Circadian rhythm :

Dominant pacemaker AKA master clock : Suprachiasmatic nucleus (SCN).

Function : Synchronize 24 hr bodily functions.

2. Regulation of food intake :

Arcuate nucleus (integration center)

- Alpha melanocyte stimulating hormone(MSH) Cocaine amphetamine regulated Transcript (CAR) → Ventromedial nucleus (satiety center) → ↓ food intake
- Neuropeptide Y (NPY) Agouti related peptide (AGRP) → Lateral nucleus → Orexin neurotransmitter → ↑ food intake

3. Regulation of body temperature :

Hot temperature → Anterior nucleus → Sweating. Vasodilation. Thirst (Osmoreceptor).

Cold temperature → Posterior nucleus → Shivering. Vasoconstriction. ↑ urinary frequency.

4. Regulation of sleep & wakefulness :

Awake/arousal state :

- Lateral nucleus : Orexin.
- Posterior nucleus : Histamine.

Sleep centre : Ventrolateral pre-optic nucleus.

5. Role in memory :

- mammillary body.

Ask Feedback

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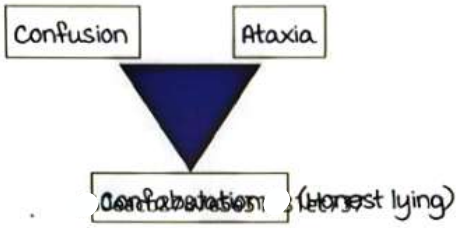


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Applied aspect :

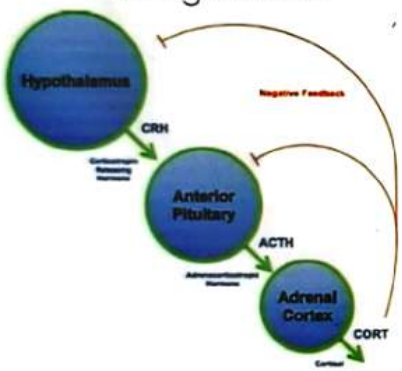
Chronic alcoholism → memory impaired (D/t degeneration of mammillary bodies).

Presentation :



Wernicke-Korsakoff syndrome

6. Endocrine functions :



Hypothalamopituitary axis

Inhibiting hormone :
Dopamine (Prolactin inhibiting factor)
↓
Inhibits prolactin release from anterior pituitary

Paraventricular nucleus
↓
major : Oxytocin.
minor : ADH.

Supraoptic nucleus
↓
major : Vasopressin (ADH).
minor : Oxytocin.

Stored & released in posterior pituitary

CLINICAL ASPECT :

- In healthy individual, cerebral cortex
↓
Hypothalamus (Dorsomedial nucleus)
↓
Sham rage not seen

- On decortication (Removal of cerebral cortex)
↓
Sham rage



Sham Rage (Dorsomedial nucleus)
(Removal of hypothalamus : Abolish sham rage)

Feedback

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LEARNING AND MEMORY

----- Active space ----

Cerebral Hemispheres

00:03:25

TYPES :

Features	Left	Right
Old name	Dominant hemisphere • 96 % of (R) handed • 70 % of (L) handed	Non dominant hemisphere
New name	Categorical hemisphere	Representational hemisphere
Functions	• Language. • Speech. • Learning & memory.	• Visuo-spatial representation. • Creativity. • Musical ability.
Thinking	Logic based	Intuition based
Lesions	• Aphasias. • Dyscalculias.	• Astereognosis. • Agnosia. • Hemispatial neglect.

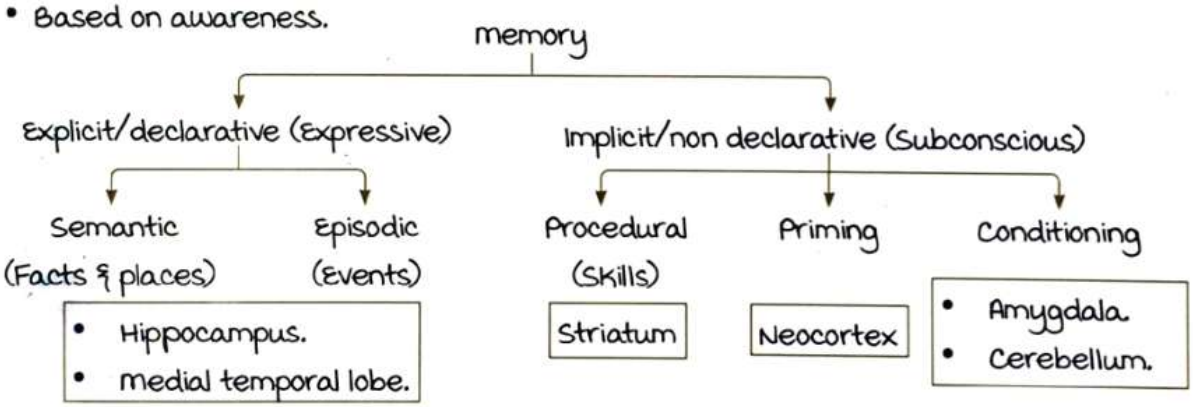
Memory

00:12:50

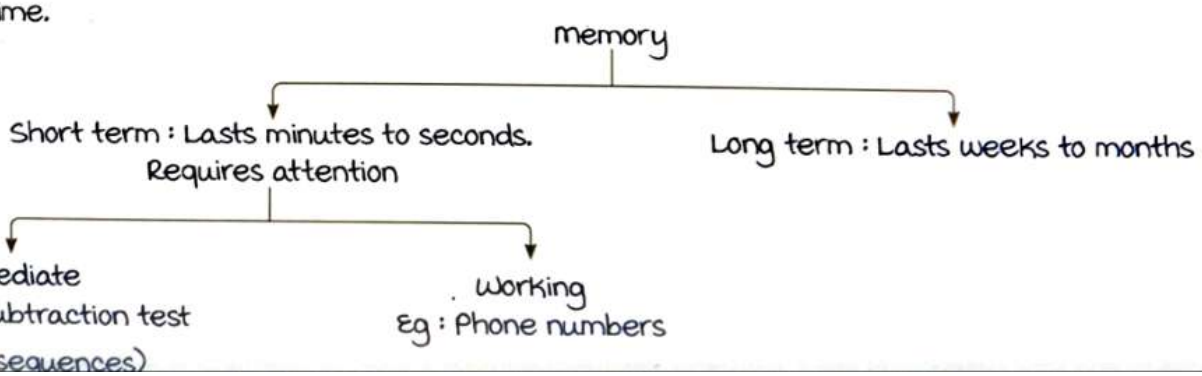
Memory engram : Collection of memories.

Types :

- Based on awareness.



- Based on time.



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LONG TERM MEMORY :

Synaptic plasticity :

- Modification of strength of synaptic transmission.
- Based on activity.

Types

- Potentiation
  - Post tetanic potentiation
  - Long term potentiation (LTP)
  - Sensitization
- Depression
  - Long term depression (LTD)
  - Habituation

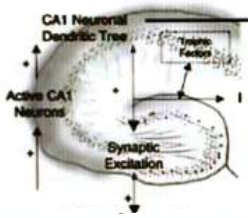
Long term potentiation (LTP) :

Occurs in hippocampus/Cornu ammonis (CA)/Ammon's horn.

- Fn : Short term memory —Consolidation/LTP→ Long term memory



Hippocampus



Cornu ammonis I neurons : memory neurons

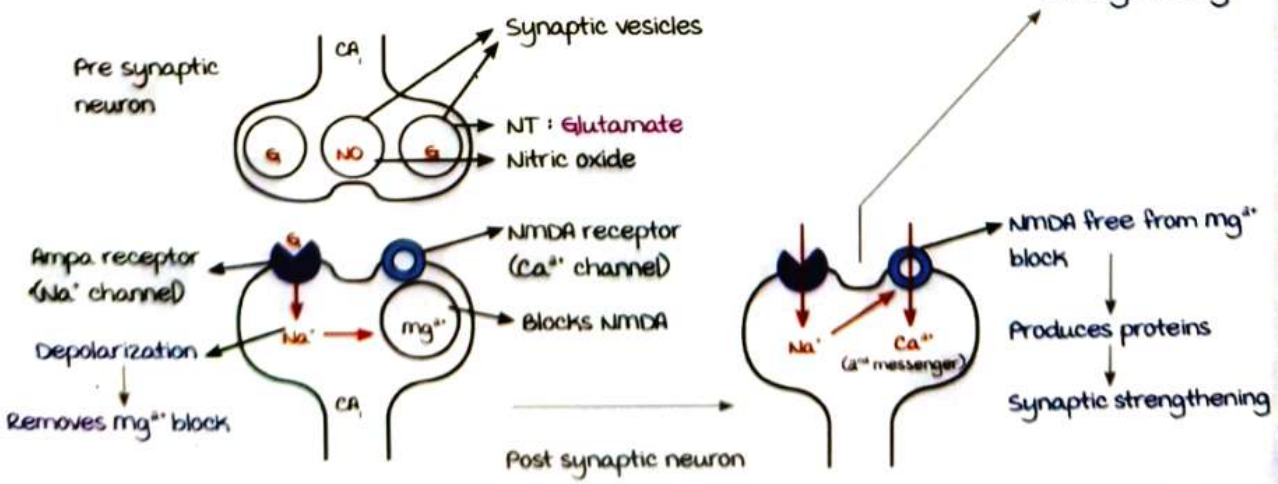
- Pathway

Sensory inputs
↓
Perforant pathways
↓
Entorhinal cortex

- Direct pathway → CA I neurons
- Indirect pathway → Dentate part (Hippocampus) → CA 3 neurons → Schaffers collateral → CA I neurons

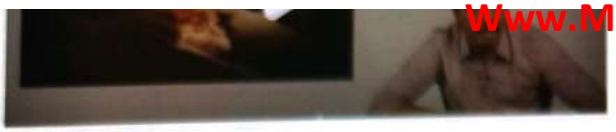
- Mechanism :

Hebb's rule : Repeated activation of hippocampal synapses → Synaptic strengthening.



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- Applied aspect : Lesions in hippocampus → Anterograde amnesia.
  - Inability to form new long term memories.
  - Intact memories : Recent, short term, procedural, past.

----- Active space -----

Long term depression (LTD) :

- Occurs in cerebellum
- Significance :

  LTD → Faulty synapses : Depressed → Correct synaptic connections : Established

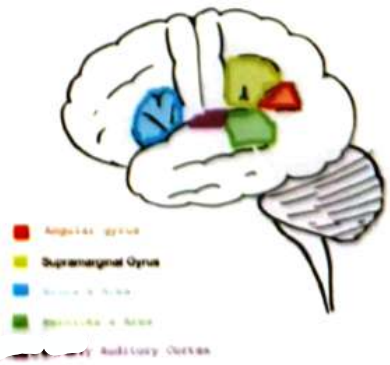
OTHER AREAS :

1. Mamillary bodies :
   - Function : Recollective memory.
   - Degeneration → Wernicke Korsakoff psychosis (Risk factor : Alcoholics).
2. Thalamus : Anterior nucleus.
   - Function : Recent memories.
3. Basal forebrain : Nucleus basalis of meynert.
   - Neurotransmitters : Acetylcholine.
   - Degeneration → Alzheimer's disease.
4. Amygdala :
   - Function : Associates memory with emotions.
5. Entorhinal cortex :
   - Function : Associates memory with smell.
6. Corpus callosum :
   - Function : Hemispheric transfer of memory.

Speech

00:46:50

Area	Brodmann area	Function	Disorder
Broca's area (Inferior frontal cortex)	44	Motor speech area : (vocalisation) • Speech output. • Verbal expression. • Word formation.	Non fluent aphasia : • Speech output ↓. • Repetition : Impaired.
Wernicke's area (Superior temporal gyrus)	22	Sensory speech area : • Understanding. • Comprehension.	Fluent aphasia : • Speech output ↑. • Comprehension : Lost.
Arcuate fasciculus	-	Connects broca's & wernicke's area.	Conduction aphasia
Angular gyrus	39	Integrates vision & speech.	Anomic aphasia (Inability to name an object after seeing)



Speech areas

Feedback